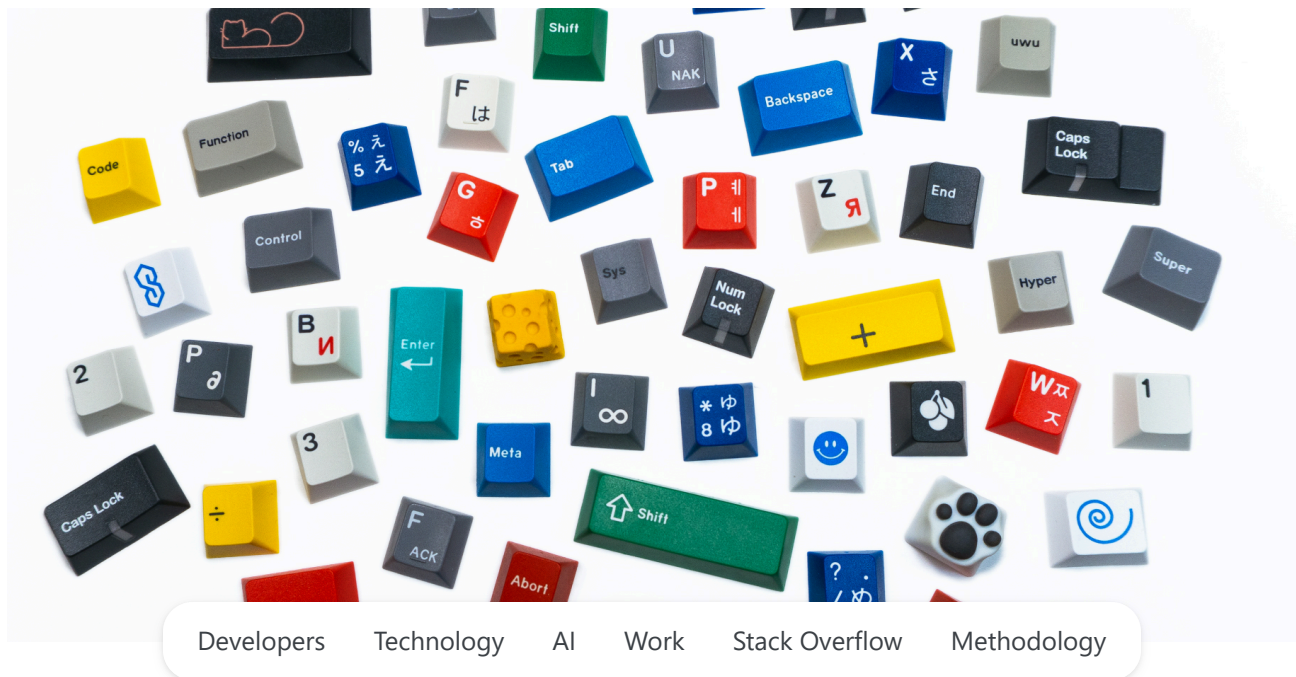




Products

The 2025 Developer Survey is the definitive report on the state of software development. In its fifteenth year, Stack Overflow received over 49,000+ responses from 177 countries across 62 questions focused on 314 different technologies, including new focus on AI agent tools, LLMs and community platforms. This annual Developer Survey provides a crucial snapshot into the needs of the global developer community, focusing on the tools and technologies they use or want to learn more about.

[Developers](#) [Technology](#) [AI](#) [Work](#) [Stack Overflow](#) [Methodology](#)[Developers](#) → [Education & Experience](#)

More than one third of respondents use AI-enabled tools to learn AI this year

Yes, I learned how to  **36.3%**

Yes, I learned how to  **31%**

No, I learned someth  **11.7%**

No, I didn't spend tim  **10.8%**

No, I learned someth  **10.2%**

Over 36% of respondents learned how to use AI-enabled tools for their job or to advance their career in the last year.

[Learn to code for AI →](#)

[Developers](#) → [Education & Experience](#)

Most developers have been coding for 10+ years

1 to 5 years  **13.9%**

6 to 10 years  **21.1%**

11 to 15 years  **15.6%**

16 to 20 years  **11.8%**

21 to 30 years  **14.6%**

31 to 40 years  **7%**

41 to 50 years  **3.1%**

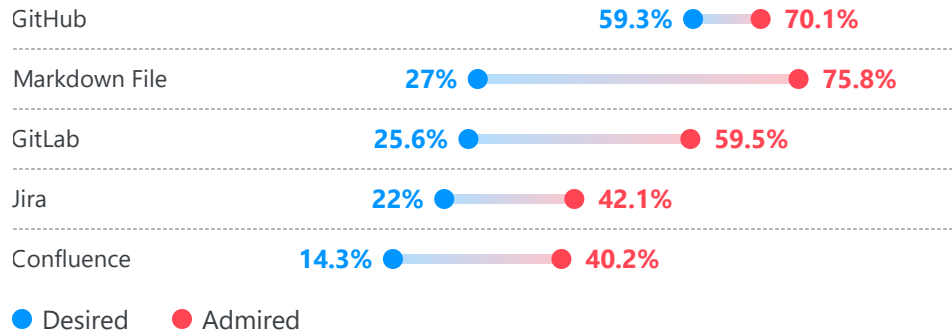
More than 50 years  **0.5%**

35% of developers responding to the Developer Survey this year have been coding less than 10 years.

[Years coding →](#)

[Technology](#) → [Admired and Desired](#)

GitHub is a more desirable collaboration tool than Jira this year

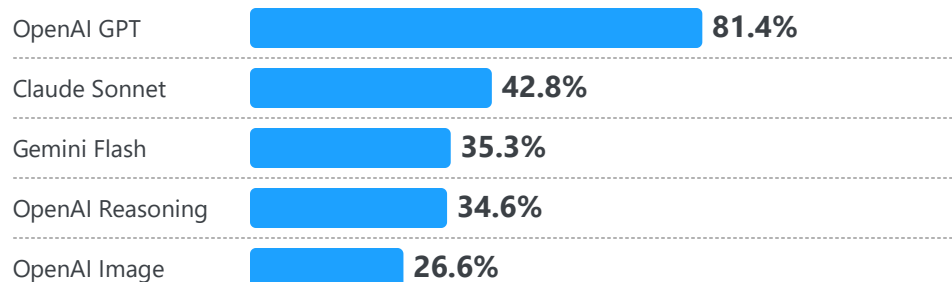


Jira steps down as the most desired tool for code documentation and collaboration and the new top desired tool is GitHub. Markdown continues to be the most admired sync tool for the third year.

[Code documentation and collaboration tools →](#)

[Technology](#) → [Most popular technologies](#)

A vast majority of developers indicating they worked with OpenAI GPT models in the past year

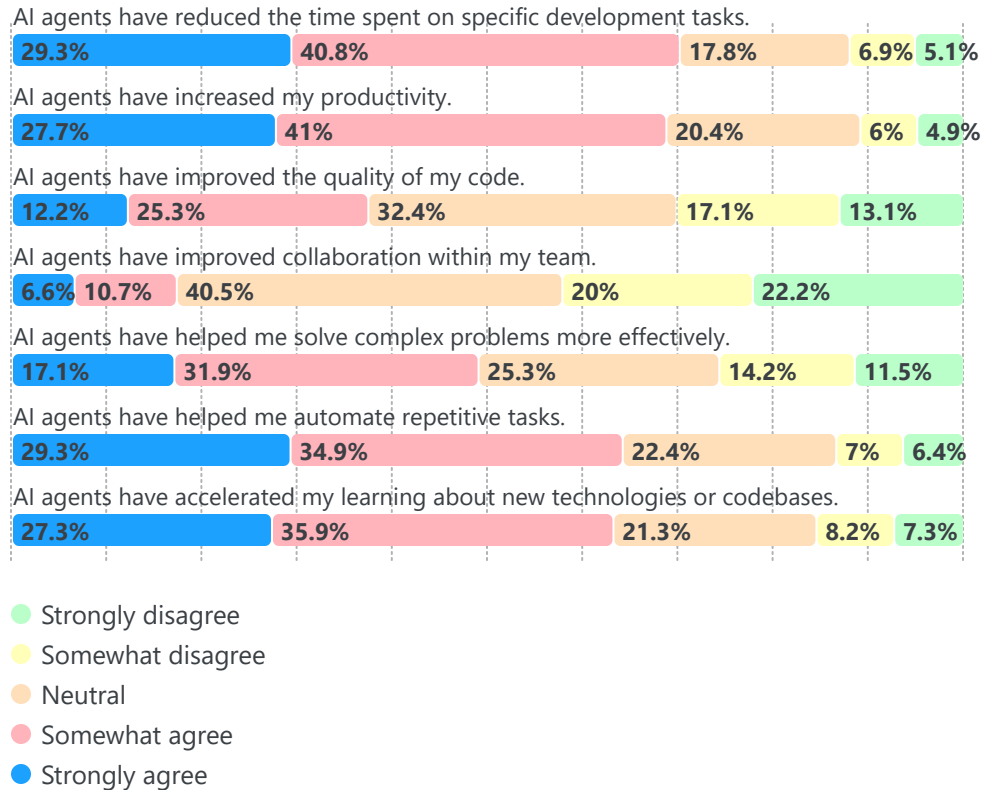


OpenAI's GPT models top the large language model list with 82% of developers indicating they used them for development work in the past year. Anthropic's Claude Sonnet models are used more by professional developers (45%) than by those learning to code (30%).

[Large language models →](#)

[AI](#) → [AI Agents](#)

69% of AI agent users agree AI agents have increased productivity



The most recognized impacts are personal efficiency gains, and not team-wide impact. Approximately 70% of agent users agree that agents have reduced the time spent on specific development tasks, and 69% agree they have increased productivity. Only 17% of users agree that agents have improved collaboration within their team, making it the lowest-rated impact by a wide margin.

Impacts of AI agents →

[AI](#) → [Sentiment and usage](#)

84% of respondents are using AI tools this year

Yes, I use AI tools da  47.1%

Yes, I use AI tools we  17.7%

Yes, I use AI tools mo  13.7%

No, but I plan to soo  5.3%

No, and I don't plan  16.2%

84% of respondents are using or planning to use AI tools in their development process, an increase over last year (76%). This year we can see 51% of professional developers use AI tools daily.

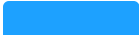
AI tools in the development process →

AI → [Developer tools](#)

66% of developers are frustrated with AI solutions that are almost right

AI solutions that are  66%

Debugging AI-gener  45.2%

I don't use AI tools r  23.5%

I've become less con  20%

It's hard to understa  16.3%

Other (write in):  11.6%

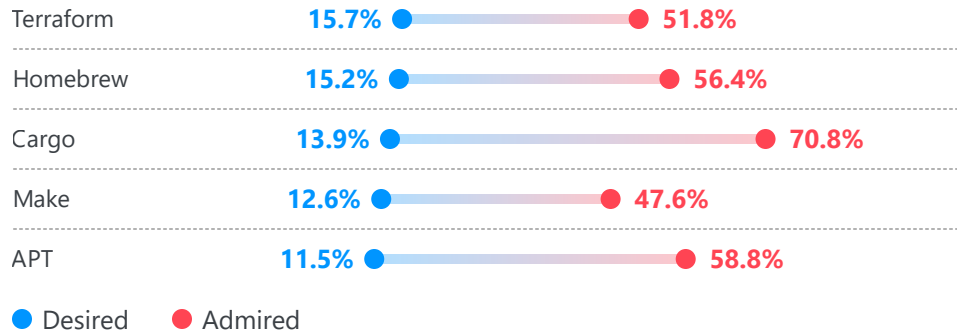
I haven't encountere  4%

The biggest single frustration, cited by 66% of developers, is dealing with "AI solutions that are almost right, but not quite," which often leads to the second-biggest frustration: "Debugging AI-generated code is more time-consuming" (45%)

AI tool frustrations →

Technology → [Admired and Desired](#)

Cargo is the most admired cloud development and infrastructure tool this year

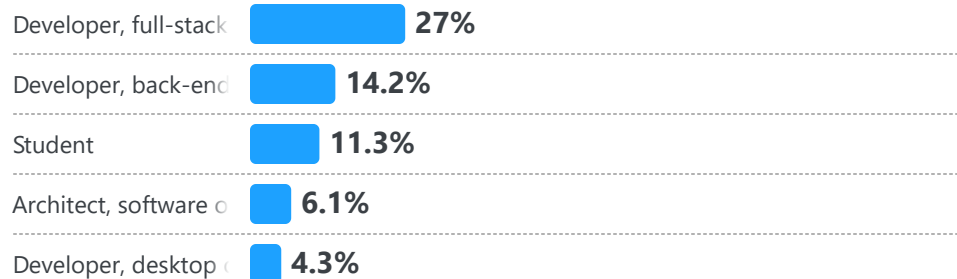


Rust's growth is directly tied to the success of its build tool and package manager, Cargo, which is the most admired (71%) cloud development and infrastructure tool this year.

Cloud development →

[Developers](#) → [Developer roles](#)

Architect is the fourth top role for developers this year

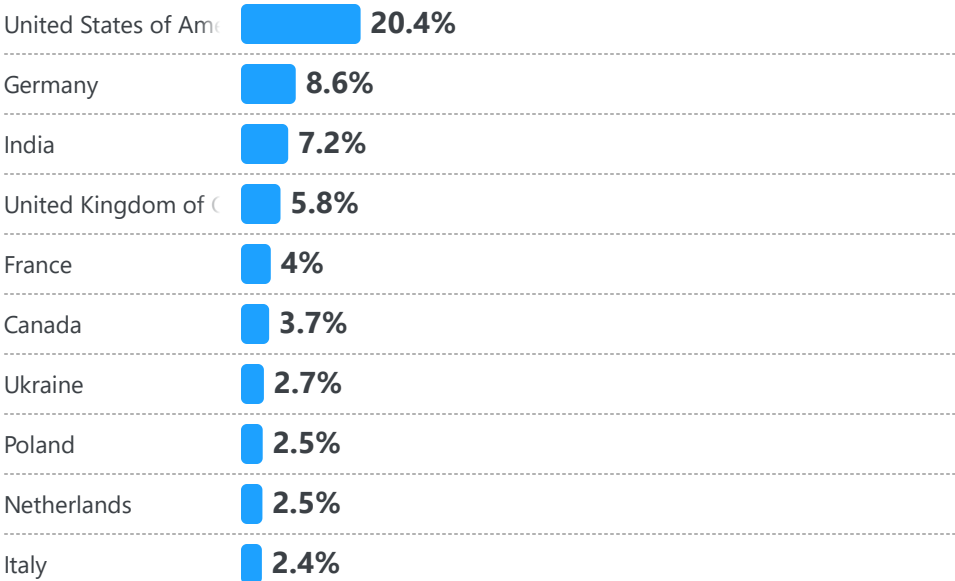


Architect is a new role we added to the survey this year and is the fourth most popular role for respondents.

Role →

[Developers](#) → [Profile](#)

USA, Germany and India are top countries responding to this year's survey

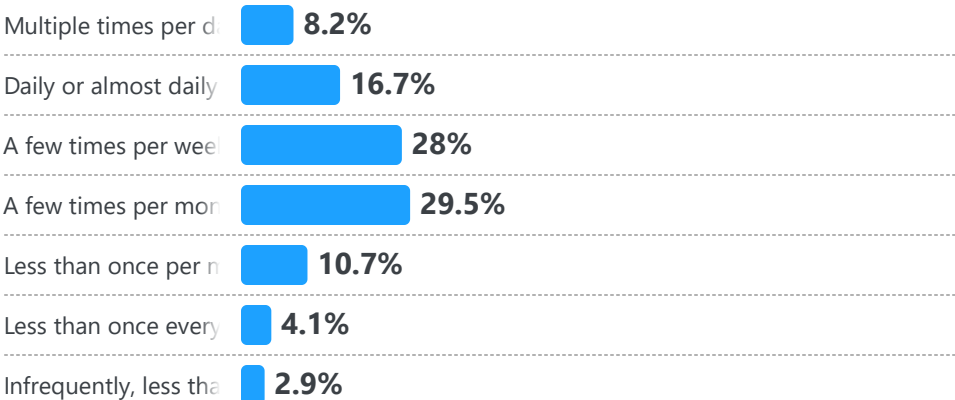


Ukraine and France swapped places this year compared to last, placing France in the top 5 list of responding countries.

Country →

[Stack Overflow](#) → [Stack Overflow site use](#)

Stack Overflow is a destination developers visit frequently

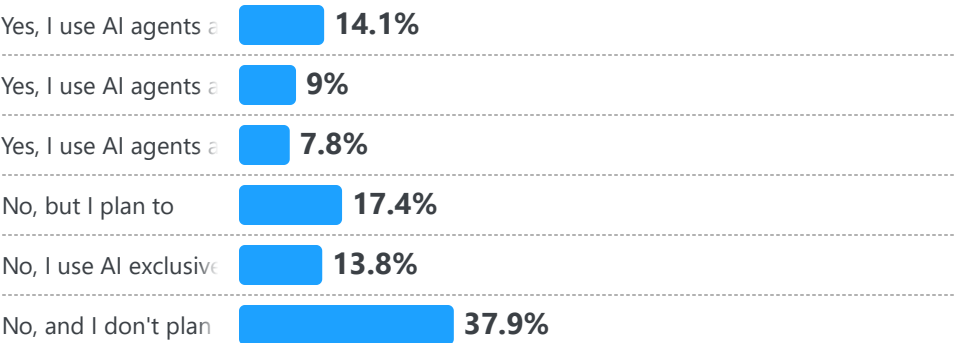


Stack Overflow is a frequent destination for information. A strong majority (82%) visit at least a few times per month, with 25% visiting daily or more often.

Frequency of visiting Stack Overflow →

AI → AI Agents

A majority of developers don't use AI agents



AI agents are not yet mainstream. A majority of developers (52%) either don't use agents or stick to simpler AI tools, and a significant portion (38%) have no plans to adopt them.

AI agents →

Work → Technology purchases

Privacy, pricing and better alternatives are top reasons developers turn their back on a technology

Detraction	Overall Rank	Median Rank	Mode Rank
Security or privacy concerns	1	3	1
Prohibitive pricing	2	4	1
Availability of better alternatives	3	4	1
Poor usability	4	4	4

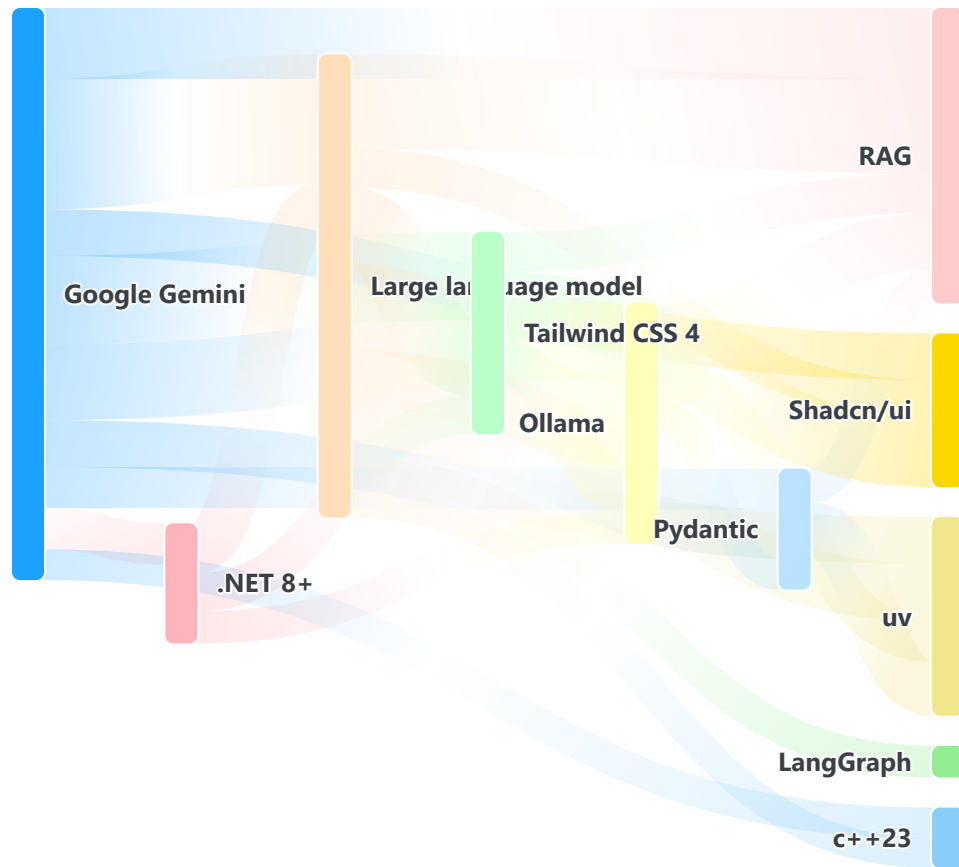
Detraction	Overall Rank	Median Rank	Mode Rank
Inefficient or time-costly	5	5	5
Outdated or obsolete technology or features	6	5	8
Ethical concerns	7	6	8
Lack of or sub-par API	8	6	8
Lack of AI or AI agents	9	9	9
Other	10	10	10

The reasons to reject a technology are nearly universal. The top three deal-breakers for all developers are security or privacy concerns (Rank 1), prohibitive pricing (Rank 2), and the availability of better alternatives (Rank 3). The lack of AI is the least important factor (Rank 9).

[How you lose interest in tech tools →](#)

[Technology](#) → [Worked with vs. want to work with](#)

Developers at all levels are exploring the evolving AI landscape through Stack Overflow



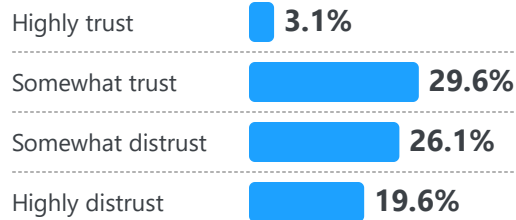
Most professional developers who indicated they used Google Gemini last year, are interested in other AI-oriented subjects like "Large Language Model" or "RAG", and tools like "Ollama".

This pattern holds true for both Professional Developers and those Learning to Code, showing that developers at all levels are actively exploring the rapidly evolving AI landscape rather than committing to a single tool or platform.

Stack Overflow tags →

[AI](#) → [Developer tools](#)

More developers actively distrust the accuracy of AI tools than trust it

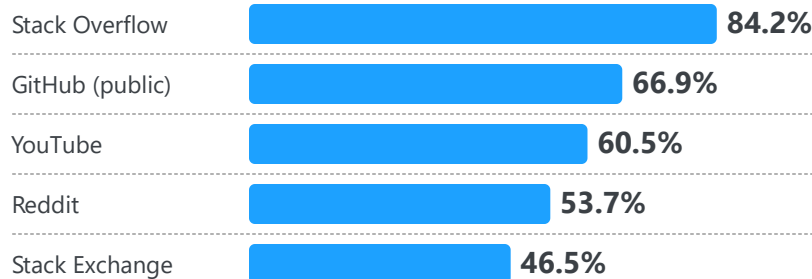


More developers actively distrust the accuracy of AI tools (46%) than trust it (33%), and only a fraction (3%) report "highly trusting" the output. Experienced developers are the most cautious, with the lowest "highly trust" rate (2.6%) and the highest "highly distrust" rate (20%), indicating a widespread need for human verification for those in roles with accountability.

Accuracy of AI tools →

Technology → Most popular technologies

Respondents learning to code use YouTube for community more than professional developers

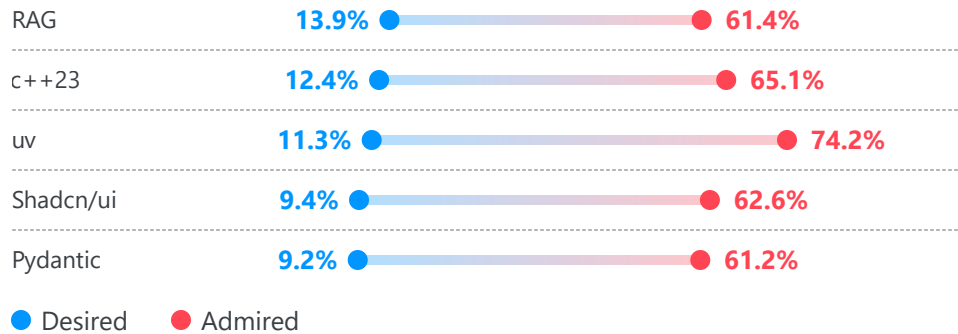


Respondents learning to code use Youtube for community more than professional developers (70% vs. 60%).

Community platforms →

Technology → Admired and Desired

uv is the most admired SO tag technology this year

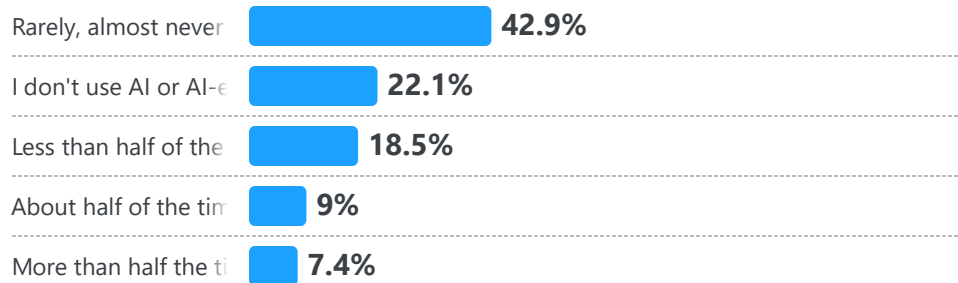


uv is a Python package manager built in Rust; need we say more about why this is the most admired (74%) SO tag technology this year?

[Stack Overflow tags](#) →

[Stack Overflow](#) → [Participation and Feedback](#)

Stack Overflow is becoming a new resource for developers that need to solve AI-related issues

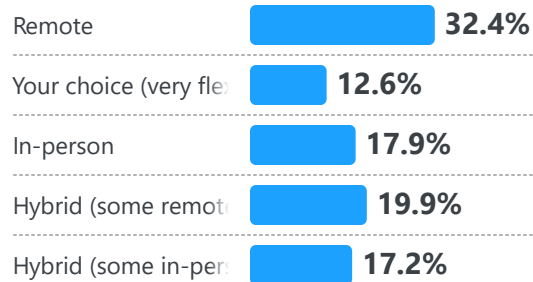


Developers turn to Stack Overflow for human-verified, trusted knowledge. About 35% of developers report that their visits to Stack Overflow are a result of AI-related issues at least some of the time.

[Frictions on Stack Overflow](#) →

[Work](#) → [Employment](#)

Nearly one third of developers are working remote this year

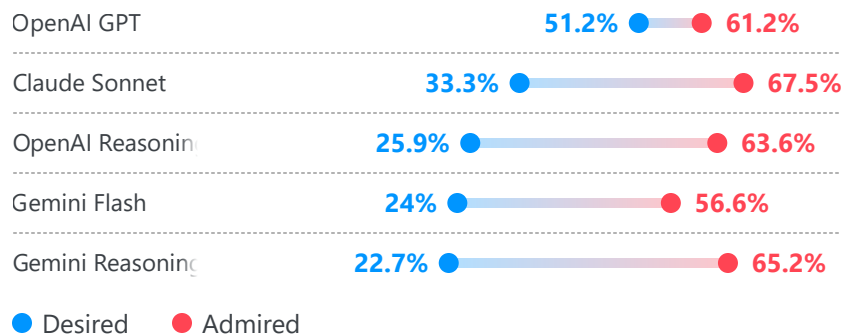


Of the top-reporting countries in this year's survey, the US has the highest number of developers working remotely (45%). 21% of developers in Germany say the choice to go into the office or work remotely is completely up to them.

Work environment →

Technology → Admired and Desired

Claude Sonnet is the most admired AI model



Anthropic's Claude Sonnet is the most admired LLM this year (behind Gemini Reasoning) and second most desired (33%).

Large language models →

Work → Job Satisfaction

One in four developers are happy at their current job

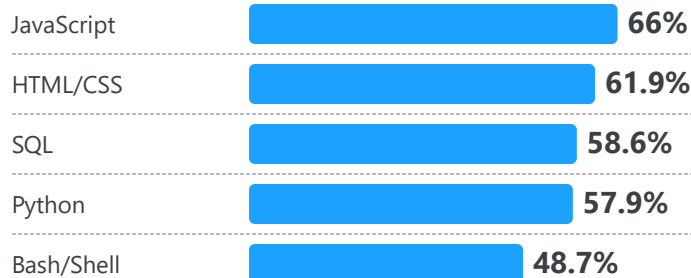


More developers are happy at work this year (24% vs. 20% last year). This is likely related to that pay bump in the data for certain roles this year.

[Job satisfaction](#) →

[Technology](#) → [Most popular technologies](#)

Python adoption grew in 2025

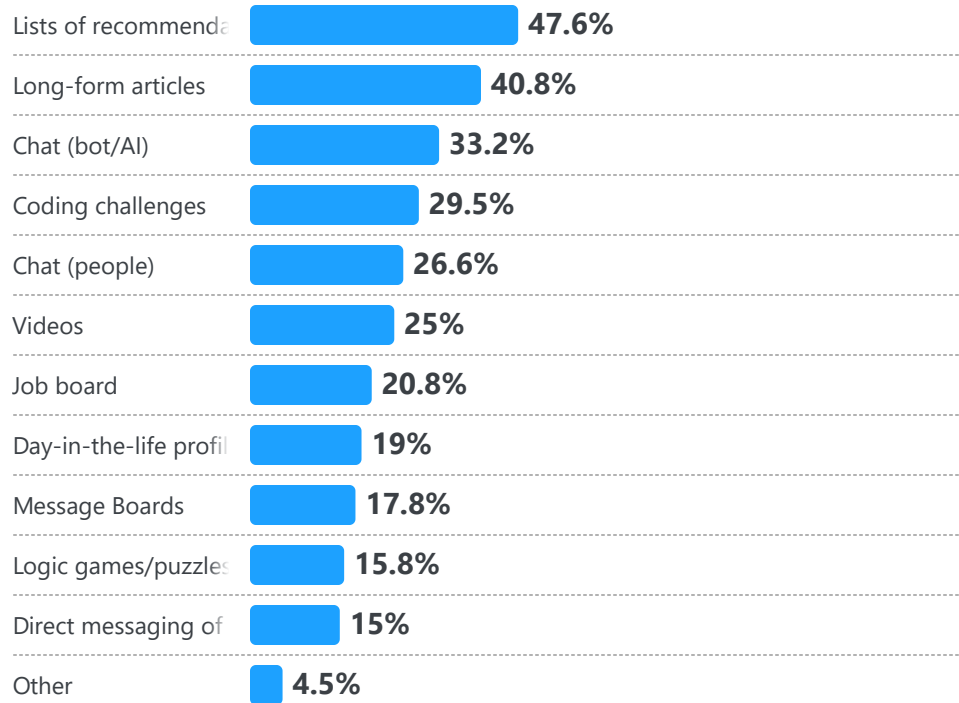


After more than a decade of steady growth, Python's adoption has accelerated significantly. It saw a 7 percentage point increase from 2024 to 2025; this speaks to its ability to be the go-to language for AI, data science, and back-end development.

[Programming, scripting, and markup languages](#) →

[Stack Overflow](#) → [Participation and Feedback](#)

Younger developers want developer content with social or interactive formats

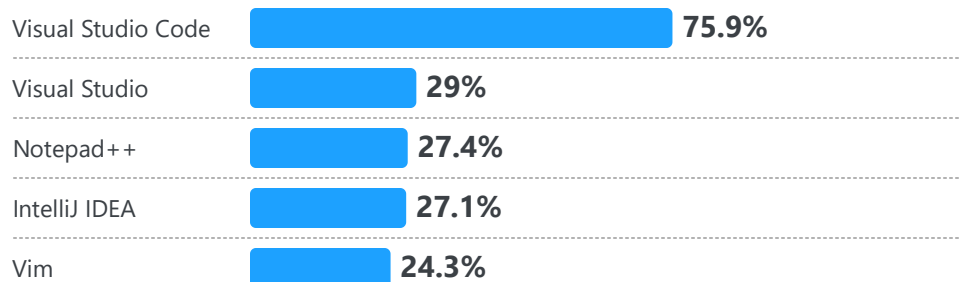


While all age groups want lists and articles, younger developers show a significantly higher interest in more social and interactive formats. For example, 37% of 18-24 year olds want "Chat (people)", compared to only 20% of 55-64 year olds. Similarly, 39% of the youngest cohort want "Coding challenges," also aligning with a motivation to skill up.

How do you choose to find relevant developer content? →

[Technology](#) → [Most popular technologies](#)

Visual Studio and Visual Studio Code both maintained their top spots for developer environments used for the fourth year

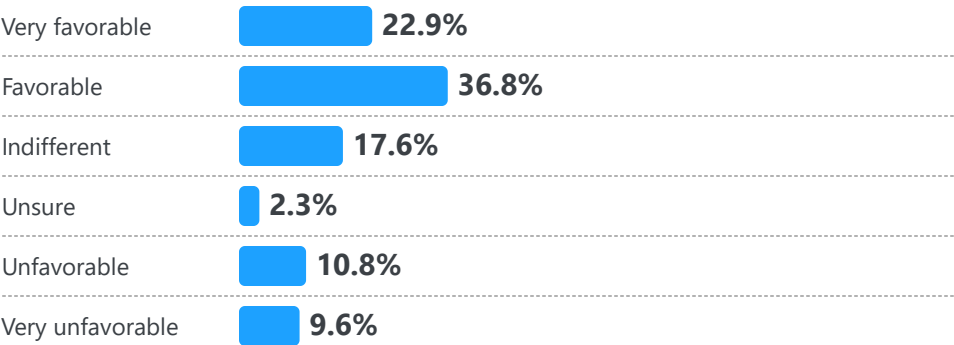


Subscription-based, AI-enabled IDEs weren't able to topple the dominance of Visual Studio and Visual Studio Code this year. Both maintained their top spots for the fourth year while relying on extensions as optional, paid AI services.

Dev IDEs →

AI → Sentiment and usage

Positive sentiment to AI tools has decreased in 2025



Conversely to usage, positive sentiment for AI tools has decreased in 2025: 70%+ in 2023 and 2024 to just 60% this year. Professionals show a higher overall favorable sentiment (61%) than those learning to code (53%).

AI tool sentiment →

Survey index

1. Developers	2. Technology	3. AI	4. Work	5. Stack Overflow
1.1. Profile	2.1. Most popular technologies	3.1. Sentiment and usage	4.1. Employment	5.1. Stack Overflow use
Coding			Employment status	Have an account on Stack Overflow
Age	Programming, scripting, and markup languages	AI tools in the development process	Employment additional status	How long had an account on Stack Overflow
Country				

1.2.
Education
&
Experience

Educational attainment
Years coding
Learn to code
How did you learn to code?
Learn to code for AI
How did you learn to code for AI?

1.3.
Developer
roles

Role
Work experience
Industry

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Cloud development
Web frameworks and technologies
Dev IDEs
Stack Overflow tags
Community platforms
Large language models
Code documentation and collaboration tools
Computer operating systems

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and Desired

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Cloud development
Web frameworks and technologies
Dev IDEs
Stack Overflow tags
Community platforms
Large language models
Code documentation and

AI tool
sentiment

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AI in the development workflow
AI workflow and tool satisfaction

AI tool
frustrations

AI and
humans in
the future

Vibe coding

3.3. AI
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AI agents affect on work productivity
AI agent uses at work
AI agent uses for general purposes
Impacts of AI agents
Challenges with AI agents
AI Agent data storage tools

Work
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IC or PM

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Company size

4.3. Salary

Salary by
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Salary and
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developer type

4.4.
Technology
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Influence on
technology
purchases

What makes you
a fan of specific
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How you lose
interest in tech
tools

Count of tools
used to work

Count of tools
used to code
outside of work

4.5. Job
Satisfaction

Looking for a
new role

Job satisfaction

Ranked
attributes
contributing to

Stack Over

Frequency
Stack Over

5.2.
Particip
and Fee

Frequency
participati
Stack Over

How do yo
to find rele
developer

How would
to use Stac
Overflow?

Feel like p
Stack Over
communit

Frictions o
Overflow

collaboration tools	AI Agent orchestration tools	or detracting from job satisfaction
2.3. Worked with vs. want to work with	AI Agent observability and security	Is AI a threat to your job?
Programming, scripting, and markup languages	AI Agent out-of-the-box tools	
Databases		
Cloud development		
Web frameworks and technologies		
Dev IDEs		
Stack Overflow tags		
Community platforms		
Large language models		
Code documentation and collaboration tools		



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